Digital Preservation of Cultural Heritage using XR Technologies

Krémer Aliz
Hősi Rezső Rudolf
Oláh Sára

ELTE

Executive Summary

// Rezső innen The project "Digital Preservation of Cultural Heritage using XR Technologies" aims to revolutionize how we interact with and preserve history. By leveraging Extended Reality (XR)—encompassing Virtual, Augmented, and Mixed Reality—this project focuses on creating high-fidelity digital twins of endangered cultural heritage sites. The proposal addresses the urgent need for non-destructive preservation and accessible education. Key outcomes include a standardized framework for digital heritage reconstruction, a cloud-based XR platform for remote exploration, and specialized tools for historical researchers. This summary highlights our interdisciplinary approach combining computer science with digital humanities to ensure that cultural artifacts remain accessible for future generations despite physical decay or environmental threats.

Technical and Scientific Description of the Project

1 The Project Topic and its Practical Relevance

// Sári innen The project focuses on the digital preservation and accessibility of cultural heritage sites using XR technologies. By integrating 3D scanning, photogrammetry, and immersive VR/AR environments, it enables the creation of digital twins of endangered monuments.

Two major challenges are addressed:

- Physical degradation caused by mass tourism and environmental factors
- Limited accessibility to remote or fragile heritage sites

The project operates across Mediterranean, Central-Eastern European, and Atlantic regions, combining technological innovation with cultural heritage expertise.

2 Project Contribution Beyond the State of the Art

Recent advances in XR technologies and digital twin systems have opened new possibilities in cultural heritage preservation. However, existing solutions are often limited to isolated case studies or indoor environments.

Current limitations include:

- Lack of scalability
- Limited outdoor applicability
- Weak integration between scientific documentation and public accessibility

The project advances the state of the art by:

- Applying XR technologies in complex outdoor archaeological contexts
- Integrating large-scale, multi-region datasets
- Combining scientific documentation with immersive public access

3 Project Objectives and Outcomes

The main objectives of the project are:

- Develop high-quality 3D digital representations of cultural heritage sites
- Enable virtual access through VR/AR technologies
- Reduce physical tourism pressure on vulnerable sites
- Establish standardized methodologies for digital heritage documentation
- Foster interdisciplinary collaboration between technology and humanities
- Support open science through accessible datasets and publications

4 Original and Innovative Contributions

The project introduces innovative contributions by combining XR technologies with scalable digital heritage workflows.

Key innovations include:

- Use of high-fidelity digital twins as reusable cultural assets
- Integration of XR environments with real-world heritage preservation strategies
- Standardized workflows for outdoor heritage digitization

5 Inter-, Multi-, or Trans-disciplinary Characteristics

The project adopts an interdisciplinary approach combining computer science, archaeology, cultural heritage studies, and digital humanities.

Technological development is closely integrated with heritage expertise, ensuring both scientific accuracy and user accessibility. This approach enables effective collaboration across domains and enhances the societal relevance of the project outcomes.

Impact and Dissemination of the Project Results

1 Dissemination and Exploitation

Dissemination focuses on open science and wide accessibility:

- Open access publications (Springer Nature)
- Public datasets (3D models, scans)
- VR/AR applications for public engagement

Target audiences include:

- Researchers
- Cultural institutions
- Developers and industry stakeholders

2 Possible Applications with Market Potential

3 Improvements in Quality of Life

The project delivers impact on multiple levels:
Scientific:

- New methodologies in digital heritage
- Interdisciplinary research outputs

Technological:

- Validation of XR technologies
- Advancement of digital twin applications

Societal:

- Reduced tourism pressure
- Increased accessibility to heritage

European:

- Strengthened research networks
- Contribution to EU digital heritage strategies

4 Integration in Partner Strategies

Consortium Description

1 Project Director

2 Consortium Structure

The consortium consists of interdisciplinary partners:
Technology Partner: Arts et Métiers (CAPLAB / Laval Virtual) – XR technologies and 4D data capture
Publishing Partner: Springer Nature – Open access dissemination and scientific publishing
Mediterranean Region:

- Sapienza Università di Roma – archaeology and heritage theory
- National and Kapodistrian University of Athens – outdoor heritage and risk analysis

Central-Eastern Europe:

- Eötvös Loránd University (ELTE) – digital humanities and AI-based heritage processing

Atlantic Region:

- Sorbonne Université – data collection coordination and historical architecture

3 Research Team Leaders

4 Team Structure

5 Complementarities and Synergies

Project Management

1 Work Plan, Deliverables and Load Balancing

The project is structured into the following work packages:

A detailed timeline (Gantt chart) will define milestones and deliverables across the project duration.

2 WP1: Project Management and Coordination

3 WP2: Data collection and fieldwork

4 WP3: XR development and integration

5 WP4: XR Platform Development and UX Design

6 WP5: Overtourism Management and Virtual Flow Control

7 WP6: Community Engagement and Social Impact

8 WP7: Economic Models and Revenue Generation

Leading Partner: Springer Nature

Objectives: The primary objective of this Work Package is to establish a robust financial framework that ensures the project's self-sufficiency beyond the initial 24-month funding period. We aim to transform high-fidelity digital twins into viable economic assets for local communities and the consortium by creating diverse revenue streams.

Duration: Month 7-12

Tasks:

- **Task 7.1: Digital ticketing and subscription models (B2C):**

  Development of a tiered access system for the VR platform. This includes "Basic" (free educational access), "Premium" (high-resolution guided tours), and "Seasonal Pass" (access to all regional clusters).

- **Task 7.2: Development of a licensing framework for educational institutions and museums (B2B):**

  Designing a standardized licensing package for third-party stakeholders such as schools, universities, and international museums to use the VEP engine for their own curricula or exhibitions.

- **Task 7.3: Integrating local artisanal products and services into the virtual marketplace:**

  Creating a digital bridge between virtual visitors and local craftspeople. This involves an e-commerce layer within the XR experience allowing users to purchase physical goods from the CEE, Mediterranean, or Atlantic regions.

Deliverables:

- **Deliverable 7.1: Comprehensive Business Sustainability Plan:**

  5-year financial projection including revenue streams and operational costs.

- **Deliverable 7.2: Standardized Licensing Agreement Template:**

  Legal documentation for B2B partnerships.

Milestones:

- **Milestone 7:**

  Execution of the first commercial license agreement with a non-consortium educational entity (Month 20).

Geographical Relevance: Revenue models will be tailored to regional strengths: emphasizing the "Digital Souvenir" market in the Mediterranean and "Cultural Educational Kits" in the CEE region.

9 WP8: Testing, Validation and Pilot Actions

Leading Partner: Arts et Métiers (CAPLAB / Laval Virtual)

Objectives: This phase focuses on the technical and social validation of the developed prototypes. By deploying the platform in diverse geographic settings, we ensure the technology is resilient to varying environmental factors and user backgrounds, while measuring the actual impact on physical site preservation.

Duration: Month 4-12

Tasks:

- **Task 8.1: Implementation of Multi-Regional Pilot Events:**

  Organizing four physical and virtual "Launch Days" across the Atlantic (focusing on coastal erosion sites), Mediterranean (ancient ruins), and CEE (castles and wooden churches).

- **Task 8.2: Technical Performance and Stress Testing:**

  Evaluating rendering performance across different devices (from high-end VR headsets to mobile AR) and network bandwidth conditions.

- **Task 8.3: Comprehensive User Feedback Collection:**

  Utilizing User Experience (UX) research methods, including eye-tracking during VR sessions and qualitative focus groups with diverse age demographics to refine the interface.

Deliverables:

- **Deliverable 8.1: Pilot Evaluation Report:**

  A technical summary of the performance metrics and bug reports gathered during the pilots.

- **Deliverable 8.2: Social Impact and UX Assessment:**

  Detailed analysis of how the virtual experience shifted visitor attitudes toward physical site conservation.

Milestones:

- **Milestone 8:**

  Successful completion of all four regional pilot actions with a minimum 85% positive user satisfaction rate (Month 22).

Geographical Relevance: Testing will address specific regional challenges, such as high-glare lighting conditions for AR in Greece and hardware resilience in high-humidity Atlantic environments.

10 WP9: Sustainability, Dissemination and Communication

Leading Partner: Springer Nature

Objectives: Following the **CHARM-EU** model of long-term alliance strategy, this WP focuses on the post-project life cycle. It ensures the alliance remains sustainable through the repurposing of existing resources, the establishment of a permanent legal identity, and the wide dissemination of scientific results.

Duration: Month 1-12

Tasks:

- **Task 9.1: Scientific Dissemination and Open Science:**

  Authoring peer-reviewed publications and presenting at international heritage and technology conferences (e.g., EuroVR, Digital Heritage).

- **Task 9.2: Closing Conference and "Virtual-Europe" Exhibition:**

  A flagship event in Brussels to showcase the project results to EU policymakers, demonstrating how XR can mitigate overtourism.

- **Task 9.3: Exploitation Plan and Long-Term Alliance Strategy:**

  Developing the "Long-term strategy for sustainability of the alliance." This includes exploring the creation of a separate legal entity (e.g., an AISBL) to manage IP and maintain the digital repository beyond the pilot phase.

Deliverables:

- **Deliverable 9.1: Final Sustainability and Exploitation Plan:**

  A detailed roadmap for the alliance's activities for the next 10 years, including financial commitments from each member.

- **Deliverable 9.2: Communication Toolkit:**

  A package of branding materials and media assets for regional partners to ensure consistent messaging.

Milestones:

- **Milestone 9:**

  Final Dissemination Event and publication of the VEP White Paper on Digital Heritage Policy (Month 24).

Sustainability Strategy (Alliance Specifics):
In line with the strategy of the alliance, members are committed to action beyond the project lifecycle. This involves:

- **Repurposing existing consortium resources** to ensure continuity without requiring significant new budget requirements.
- **Investigating legal formulas** (such as a separate legal identity) to overcome bureaucratic or management barriers to long-term cooperation.
- **Leveraging existing national and European grants** to support ongoing student and researcher mobility between the Mediterranean, Atlantic, and CEE partner regions.

11 Coordination and Task Schedule

// Rezső ezt neee The project timeline spans 24 months and is structured through a phased implementation of Work Packages.

Key milestones include:

- Completion of data collection (Month 6)
- XR platform prototype (Month 12)
- Pilot testing (Month 18–22)
- Final dissemination event (Month 24)

A detailed Gantt chart defines task dependencies and scheduling across all WPs.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP 1: Management of the project	1.1	Administrative and financial management	Sapienza Università di Roma								
	1.2	Development of a Quality Assurance and Risk Management Plan									
	1.3	Data Management Plan (DMP) based on GDPR and Open Science principles									
WP2: Methodological Framework and Site Selection	2.1	Development of a vulnerability and tourism impact index.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	2.2	Multi-stakeholder analysis (involving local population, local governments, academia)									
	2.3	Finalization of site selection matrix (aspects: degree of deterioration, limited accessibility).									
WP3: High-Fidelity Data Acquisition and 3D Scanning	3.1	Laser scanning (LiDAR) and aerial photogrammetry (drone-based data collection).	Nemzeti Kapodisztriasz Athéni Egyetem (NKUA)								
	3.2	Texture optimization and material-specific digital preservation.									
	3.3	Metadata enrichment (integration of archaeological and art historical data).									

Figure 1: Project schedule (Gantt chart) of WP1, WP2 and WP3 for the 24-month period.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP4: XR Platform Development and UX Design	4.1	Development of a VR-based "Virtual Tour" engine (Unreal Engine/Unity).	Arts et Métiers (CAPLAB / Laval Virtual)								
	4.2	User Experience (UX) research and prototype testing with focus groups.									
	4.3	Development of an AR-based "On-Site Time Machine" module (ruin reconstruction via mobile application).									
WP5: Overtourism Management and Virtual Flow Control	5.1	Development of a "Virtual-First" tourism model (remote access vs. on-site restrictions).	Sorbonne Université								
	5.2	Design of digital ticketing system and "capacity monitoring" algorithms.									
	5.3	Awareness campaign about the "responsible digital tourist" concept.									
WP6: Community Engagement and Social Impact	6.1	Community workshops on the relationship between local identity and digital representation.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	6.2	Co-creation processes: integrating local legends into the XR narrative.									
	6.3	Preparation of a social impact assessment (SIA).									

Figure 2: Project schedule (Gantt chart) of WP4, WP5 and WP6 for the 24-month period.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP7: Economic Models and Revenue Generation	7.1	Digital ticketing and subscription models (B2C).	Springer Nature								
	7.2	Development of a licensing framework for educational institutions and museums (B2B).									
	7.3	Integrating local artisanal products and services into the virtual marketplace.									
WP8: Testing, Validation and Pilot Actions	8.1	Organizing pilot events in 4 countries (on-site and remote testing).	Arts et Métiers (CAPLAB / Laval Virtual)								
	8.2	Technical performance evaluation (loading speed, rendering accuracy).									
	8.3	Collect user feedback using quantitative and qualitative methods.									
WP9: Sustainability, Dissemination and Communication	9.1	Scientific publications and conference participation (Open Access).	Springer Nature								
	9.2	Organization of a closing conference and "Virtual Heritage" exhibition.									
	9.3	Development of an Exploitation Plan to adapt the technology to other regions.									

Figure 3: Project schedule (Gantt chart) of WP7, WP8 and WP9 for the 24-month period.

12 Research Infrastructure

// Aliz innen

13 Manpower Allocation

14 Budget and Partner Shares

The budget is allocated across major categories:

- Personnel costs
- Equipment and technology (XR, scanning tools)
- Travel and fieldwork
- Dissemination and publication

Detailed financial planning ensures efficient resource allocation and alignment with project objectives.

References